

Systems Thinking Capstone Project: Plastic Waste on Campus

1.Context

Plastic waste has been steadily increasing on college campuses, mainly because of the heavy use of single-use plastics. Most people choose convenience over sustainability, and the existing waste systems aren't able to handle the growing load effectively.

2. Problem Statement

Even though there are awareness campaigns, plastic waste is still rising. The main reasons are convenience-driven habits and careless disposal by students and others on campus.

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3. Key Elements

Single-use plastics, waste generation, convenience, careless disposal, cleanliness, and awareness.

4. Interconnections

The more convenient plastic is, the more people use it. This leads to more waste, which often gets disposed of carelessly. As a result, campus cleanliness drops, which in turn reduces awareness and concern about the issue.

5. Stakeholders

Students, college administration, vendors, cleaning staff, and environmental groups all play a role in this system.

6. Behavior Over Time

Plastic waste has been increasing consistently. Awareness is improving, but slowly. Cleanliness levels tend to go up and down rather than improving steadily.

7. Causal Loop Explanation

There's a reinforcing cycle where convenience keeps increasing plastic use and waste. On the other hand, there's a balancing cycle where awareness helps reduce waste, but it's not strong enough yet to counter the problem.

8. Leverage Points

Some effective ways to tackle this could be reducing the supply of single-use plastics, improving awareness campaigns, and giving incentives for better behavior.

